SCALES

DIMENSIONS OF LARGE OBJECTS MUST BE REDUCED TO ACCOMMODATE ON STANDARD SIZE DRAWING SHEET.THIS REDUCTION CREATES A SCALE OF THAT REDUCTION RATIO, WHICH IS GENERALLY A FRACTION..

SUCH A SCALE IS CALLED REDUCING SCALE

AND

THAT RATIO IS CALLED REPRESENTATIVE FRACTION

SIMILARLY IN CASE OF TINY OBJECTS DIMENSIONS MUST BE INCREASED FOR ABOVE PURPOSE. HENCE THIS SCALE IS CALLED ENLARGING SCALE. HERE THE RATIO CALLED REPRESENTATIVE FACTOR IS MORE THAN UNITY.

FOR FULL SIZE SCALE R.F.=1 OR (1:1) MEANS DRAWING & OBJECT ARE OF SAME SIZE.

Other RFs are described as

1:10, 1:100, 1:1000, 1:1,00,000

USE FOLLOWING FORMULAS FOR THE CALCULATIONS IN THIS TOPIC.

A REPRESENTATIVE FRACTION (R.F.) $= \dfrac{\text{LENGTH OF DRAWING in cm}}{\text{ACTUAL LENGTH in cm}}$

$= \sqrt{\dfrac{\text{AREA OF DRAWING in cm}^2}{\text{ACTUAL AREA in cm}^2}}$

$= \sqrt[3]{\dfrac{\text{VOLUME AS PER DRWG in cm}^3}{\text{ACTUAL VOLUME in cm}^3}}$.

B LENGTH OF SCALE (in cm) = R.F. X MAX. LENGTH TO BE MEASURED in cm.

BE FRIENDLY WITH THESE UNITS.

1 KILOMETRE = 10 HECTOMETRES = 10^5cm
1 HECTOMETRE = 10 DECAMETRES = 10^4cm
1 DECAMETRE = 10 METRES = 10^3 cm
1 METRE = 10 DECIMETRES = 10^2 cm
1 DECIMETRE = 10 CENTIMETRES = 10 cm
1 CENTIMETRE = 10 MILIMETRES

Area

1HECTARE = 10000 m^2

$1m^2 = 10^4\ cm^2$

Volume

$1m^3 = 10^6\ cm^3$

TYPES OF SCALES:

1. PLAIN SCALES (FOR DIMENSIONS UP TO SINGLE DECIMAL)
2. DIAGONAL SCALES (FOR DIMENSIONS UP TO TWO DECIMALS)
3. VERNIER SCALES (FOR DIMENSIONS UP TO TWO DECIMALS)
4. COMPARATIVE SCALES (FOR COMPARING TWO DIFFERENT UNITS)
5. SCALE OF CHORDS (FOR MEASURING/CONSTRUCTING ANGLES)

PLAIN SCALE:-This type of scale represents two units or a unit and it's sub-division.

PROBLEM NO.1:- **Draw a scale 1 cm = 1m to read decimeters, to measure maximum distance of 6 m. Show on it a distance of 4 m and 6 dm.**

PLAIN SCALE

CONSTRUCTION:-

a) Calculate R.F.= $\frac{\text{DIMENSION OF DRAWING}}{\text{DIMENSION OF OBJECT}}$

R.F.= 1cm/ 1m = 1/100

Length of scale = R.F. X max. distance

= 1/100 X 600 cm

= 6 cms

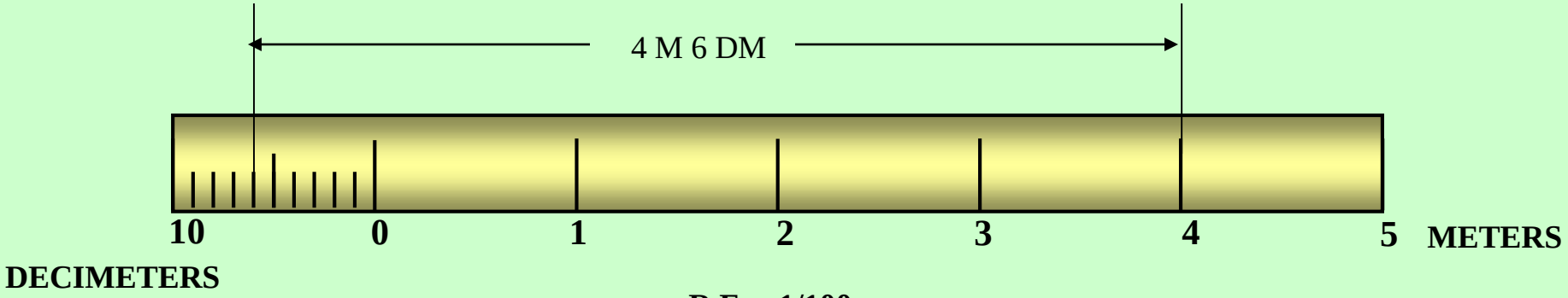
b) Draw a line 6 cm long and divide it in 6 equal parts. Each part will represent larger division unit.

c) Sub divide the first part which will represent second unit or fraction of first unit.

d) Place (0) at the end of first unit. Number the units on right side of Zero and subdivisions on left-hand side of Zero. Take height of scale 5 to 10 mm for getting a look of scale.

e) After construction of scale mention it's RF and name of scale as shown.

f) Show the distance 4 m 6 dm on it as shown.



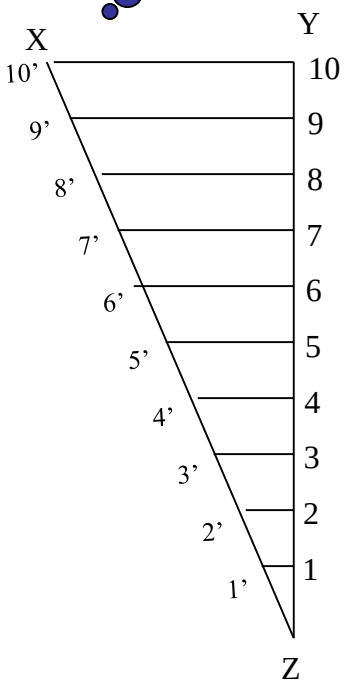
R.F. = 1/100

PLANE SCALE SHOWING METERS AND DECIMETERS.

We have seen that the plain scales give only two dimensions, such as a unit and it's subunit or it's fraction.

The diagonal scales give us three successive dimensions that is a unit, a subunit and a subdivision of a subunit.

DIAGONAL SCALE



The principle of construction of a diagonal scale is as follows.

Let the XY in figure be a subunit.
From Y draw a perpendicular YZ to a suitable height.
Join XZ. Divide YZ in to 10 equal parts.
Draw parallel lines to XY from all these divisions and number them as shown.
From geometry we know that similar triangles have their like sides proportional.

Consider two similar triangles XYZ and 7' 7Z,
we have 7Z / YZ = 7'7 / XY (each part being one unit)
Means 7' 7 = 7 / 10. XY
= 0.7 XY
:. Similarly
1' – 1 = 0.1 XY 2' – 2 = 0.2 XY
Thus, it is very clear that, the sides of small triangles, which are parallel to divided lines, become progressively shorter in length by 0.1 XY.

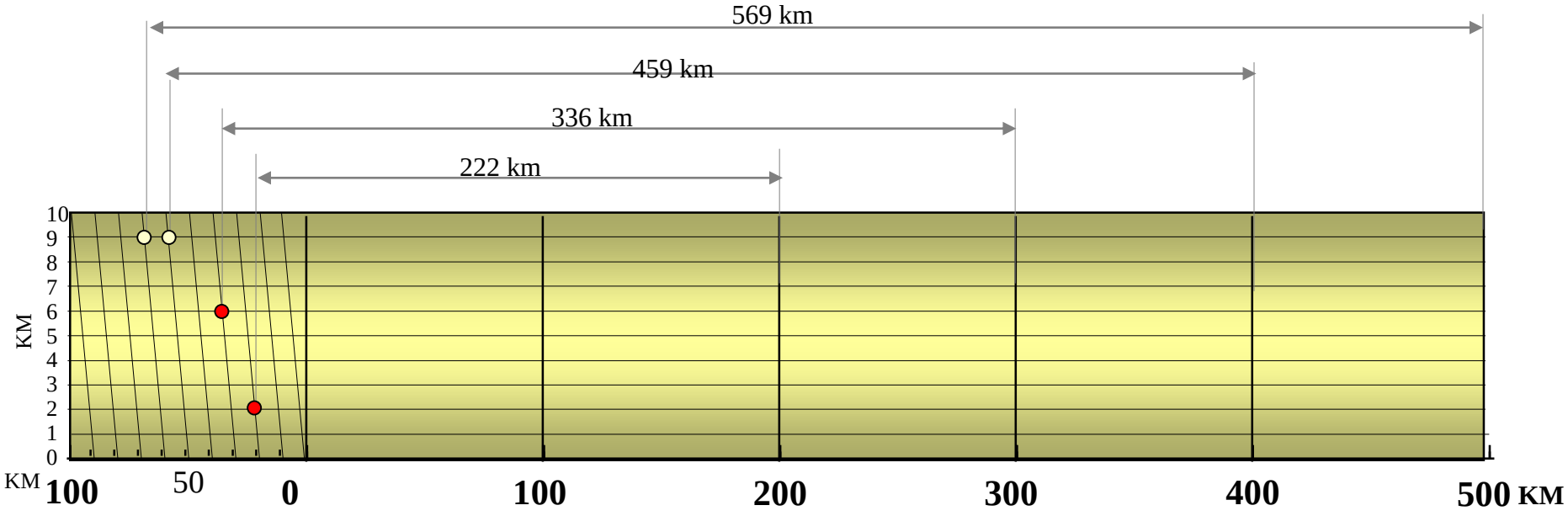
DIAGONAL SCALE

PROBLEM : The distance between Delhi and Agra is 200 km.
In a railway map it is represented by a line 5 cm long. Find it's R.F.
Draw a diagonal scale to show single km. And maximum 600 km.
Indicate on it following distances. 1) 222 km 2) 336 km 3) 459 km 4) 569 km

SOLUTION STEPS: RF = 5 cm / 200 km = 1 / 40, 00, 000

Length of scale = 1 / 40, 00, 000 X 600 X 10^5 = 15 cm

Draw a line 15 cm long. It will represent 600 km.Divide it in six equal parts.(each will represent 100 km.) **Divid**e first division in ten equal parts.Each will represent 10 km.**Draw** a line upward from left end and mark 10 parts on it of any distance. **Name** those parts 0 to 10 as shown.Join 9th sub-division of horizontal scale with 10th division of the vertical divisions. **Then** draw parallel lines to this line from remaining sub divisions and complete diagonal scale.



R.F. = 1 / 40,00,000

DIAGONAL SCALE SHOWING KILOMETERS.

PROBLEM: A rectangular plot of land measuring 1.28 hectors is represented on a map by a similar rectangle of 8 sq. cm. Calculate RF of the scale. Draw a diagonal scale to read single meter. Show a distance of 438 m on it.

SOLUTION :

1 hector = 10, 000 sq. meters

1.28 hectors = 1.28 X 10, 000 sq. meters

$= 1.28 \times 10^4 \times 10^4$ sq. cm

8 sq. cm area on map represents

$= 1.28 \times 10^4 \times 10^4$ sq. cm on land

1 cm sq. on map represents

$= 1.28 \times 10^4 \times 10^4 / 8$ sq cm on land

1 cm on map represent

$$= \sqrt{1.28 \times 10^4 \times 10^4 / 8} \text{ cm}$$

= 4, 000 cm

1 cm on drawing represent 4, 000 cm, Means RF = 1 / 4000

Assuming length of scale 15 cm, it will represent 600 m.

DIAGONAL SCALE

Draw a line 15 cm long.

It will represent 600 m.Divide it in six equal parts.

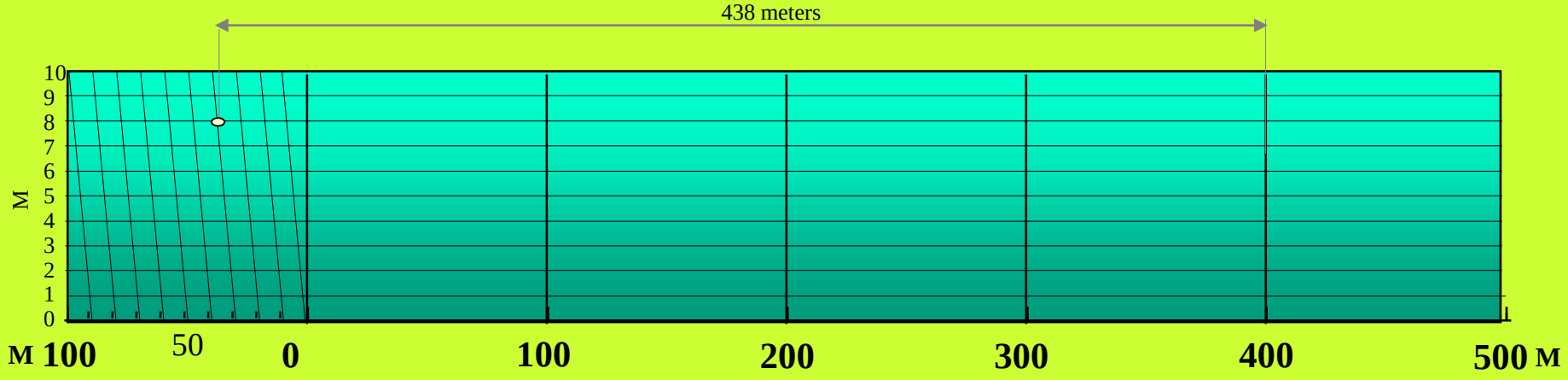
(each will represent 100 m.)

Divide first division in ten equal parts.Each will represent 10 m.

Draw a line upward from left end and mark 10 parts on it of any distance.

Name those parts 0 to 10 as shown.Join 9th sub-division of horizontal scale with 10th division of the vertical divisions.

Then draw parallel lines to this line from remaining sub divisions and complete diagonal scale.



R.F. = 1 / 4000

DIAGONAL SCALE SHOWING METERS.

COMPARATIVE SCALES:

These are the Scales having same R.F. but graduated to read different units.

These scales may be Plain scales or Diagonal scales and may be constructed separately or one above the other.

EXAMPLE:

A distance of 40 miles is represented by a line 8 cm long. Construct a plain scale to read 80 miles. Also construct a comparative scale to read kilometers upto 120 km (1 mile = 1.609 km)

SOLUTION STEPS:

***Scale of Miles*:**

40 miles are represented = 8 cm

: 80 miles = 16 cm

R.F. = 8 / 40 X 1.609 X 1000 X 100

= 1 / 8, 04, 500

CONSTRUCTION:

Take a line 16 cm long and divide it into 8 parts. Each will represent 10 miles. Subdivide the first part and each sub-division will measure single mile.

***Scale of Km*:**

Length of scale

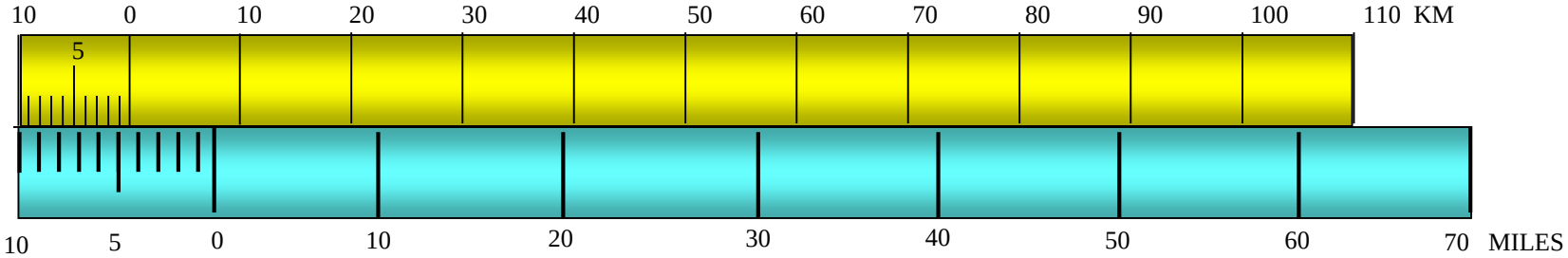
= 1 / 8,04,500 X 120 X 1000 X 100

= 14. 90 cm

***CONSTRUCTION*:**

On the top line of the scale of miles cut off a distance of 14.90 cm and divide it into 12 equal parts. Each part will represent 10 km.

Subdivide the first part into 10 equal parts. Each subdivision will show single km.



R.F. = 1 / 804500

COMPARATIVE SCALE SHOWING MILES AND KILOMETERS

EXAMPLE :

A car is traveling at a speed of 60 km per hour. A 4 cm long line represents the distance traveled by the car in two hours. Construct a suitable comparative scale up to 10 hours. The scale should be able to read the distance traveled in one minute. Show the time required to cover 476 km and also distance in 4 hours and 24 minutes.

COMPARATIVE SCALE:

SOLUTION:

4 cm line represents distance in two hours , means for 10 hours scale, 20 cm long line is required, as length of scale.This length of scale will also represent 600 kms. (as it is a distance traveled in 10 hours)

CONSTRUCTION:

Distance Scale (km)

Draw a line 20 cm long. Divide it in TEN equal parts.(Each will show 60 km)

Sub-divide 1st part in SIX subdivisions.(Each will represent 10 km)

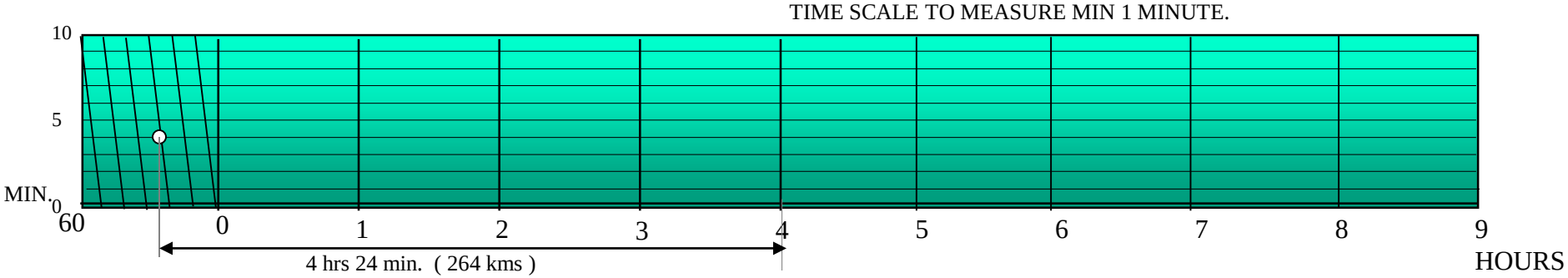
At the left hand end of the line, draw a vertical line and on it step-off 10 equal divisions of any length.

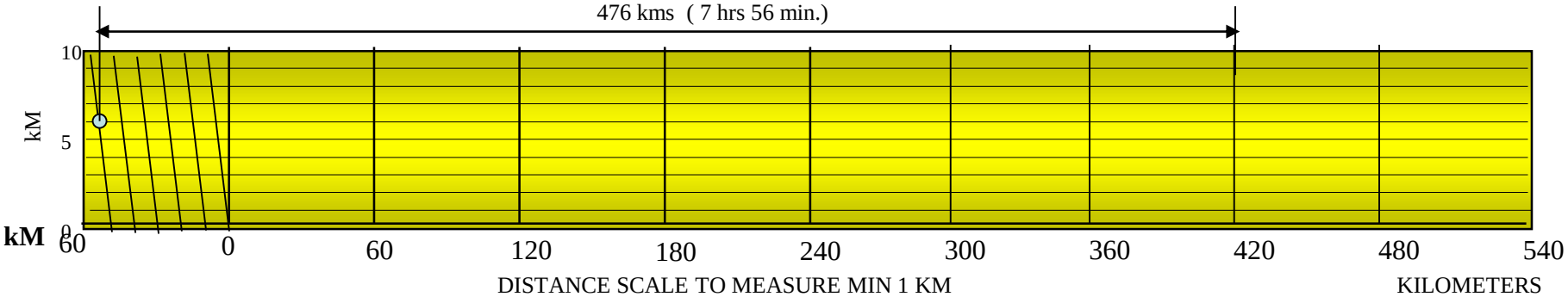
And complete the diagonal scale to read minimum ONE km.

Time scale:

Draw a line 20 cm long. Divide it in TEN equal parts.(Each will show 1 hour) Sub-divide 1st part in SIX subdivisions.(Each will represent 10 minutes) At the left hand end of the line, draw a vertical line and on it step-off 10 equal divisions of any length.

And complete the diagonal scale to read minimum ONE minute.

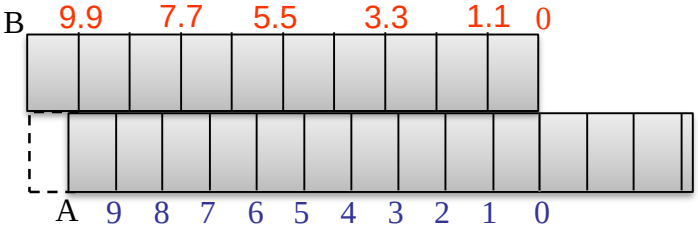




Vernier Scales:

- These scales, like diagonal scales, are used to read a very small unit with great accuracy. It consists of two parts – a primary scale and a vernier. The primary scale is a plain scale fully divided into minor divisions. As it would be difficult to sub-divide the minor divisions in ordinary way, it is done with the help of the vernier. The graduations on vernier are derived from those on the primary scale.

- Figure to the right shows a part of a plain scale in which length A-O represents 10 cm. If we divide A-O into ten equal parts, each will be of 1 cm.



- Now it would not be easy to divide each of these parts into ten equal divisions to get measurements in millimeters.

- Now if we take a length BO equal to 10 + 1 = 11 such equal parts, thus representing 11 cm, and divide it into ten equal divisions, each of these divisions will represent 11 / 10 – 1.1 cm.

- The difference between one part of AO and one division of BO will be equal 1.1 – 1.0 = 0.1 cm or 1 mm. *This difference is called Least Count of the scale.*

- The upper scale BO is the vernier. The combination of plain scale and the vernier is vernier scale.

Vernier Scale

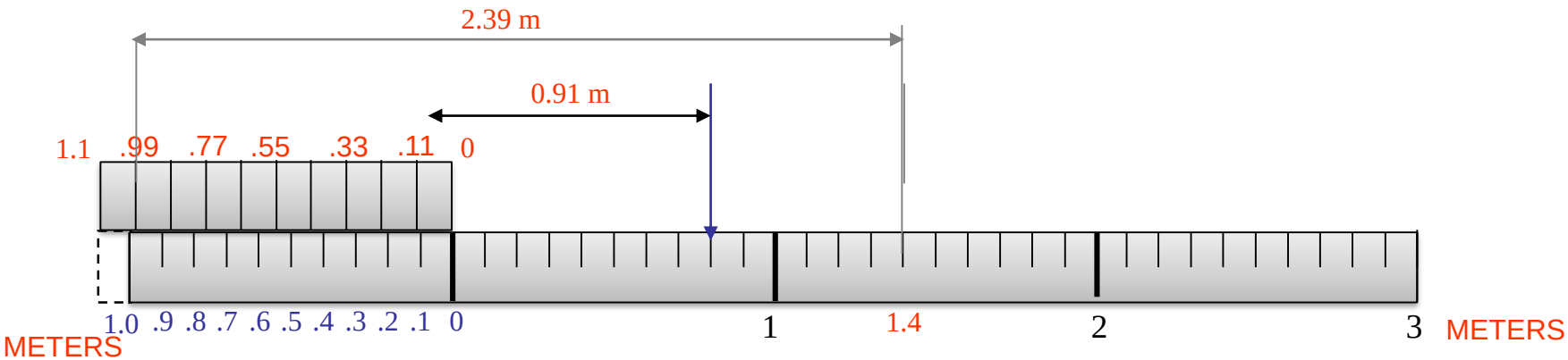
Example:
Draw a vernier scale of RF = 1 / 25 to read centimeters upto 4 meters and on it, show lengths 2.39 m and 0.91 m

SOLUTION:
Length of scale = RF X max. Distance
= 1 / 25 X 4 X 100
= 16 cm

CONSTRUCTION: *(Main scale)*
Draw a line 16 cm long.
Divide it in 4 equal parts.
(each will represent meter)
Sub-divide each part in 10 equal parts
(each will represent decimeter)
Name those properly.

CONSTRUCTION: *(vernier)*
Take 11 parts of Dm length and divide it in 10 equal parts.
Each will show 0.11 m or 1.1 dm or 11 cm and construct a rectangle
Covering these parts of vernier.

TO MEASURE GIVEN LENGTHS:
(1) For 2.39 m : Subtract 0.99 from 2.39 i.e. 2.39 - .99 = 1.4 m
The distance between 0.99 (left of Zero) and 1.4 (right of Zero) is 2.39 m
(2) For 0.91 m : Subtract 0.11 from 0.91 i.e. 0.91 – 0.11 =0.80 m
The distance between 0.11 and 0.80 (both left side of Zero) is 0.91 m



Class Assignment – LAB session

1. In a map a 36 km distance is shown by a line 45 cms long. Calculate the R.F. and construct a plain scale to read kilometers and hectometers, for max. 12 km. Show a distance of 8.3 km on it.

2. The distance between two stations is 210 km. A passenger train covers this distance in 7 hours. Construct a plain scale to measure time up to a single minute. RF is 1/200,000 Indicate the distance traveled by train in 29 minutes.

3. Draw a diagonal scale of R.F. 1: 2.5, showing centimeters and millimeters and long enough to measure up to 20 centimeters. Show distance of 13.4cm on it.

4. A motor car is running at a speed of 60 kph. On a scale of RF= 1/4,00,000 show the distance traveled by car in 47 minutes.

5. A map of size 500cm X 50cm wide represents an area of 6250 sq.Kms. Construct a Vernier scale to measure kilometers, hectometers and decameters and long enough to measure up to 7 km. Indicate on it a) 5.33 km b) 59 decameters.

Draw title block in one sheet. Each question will be evaluated for 5 marks.